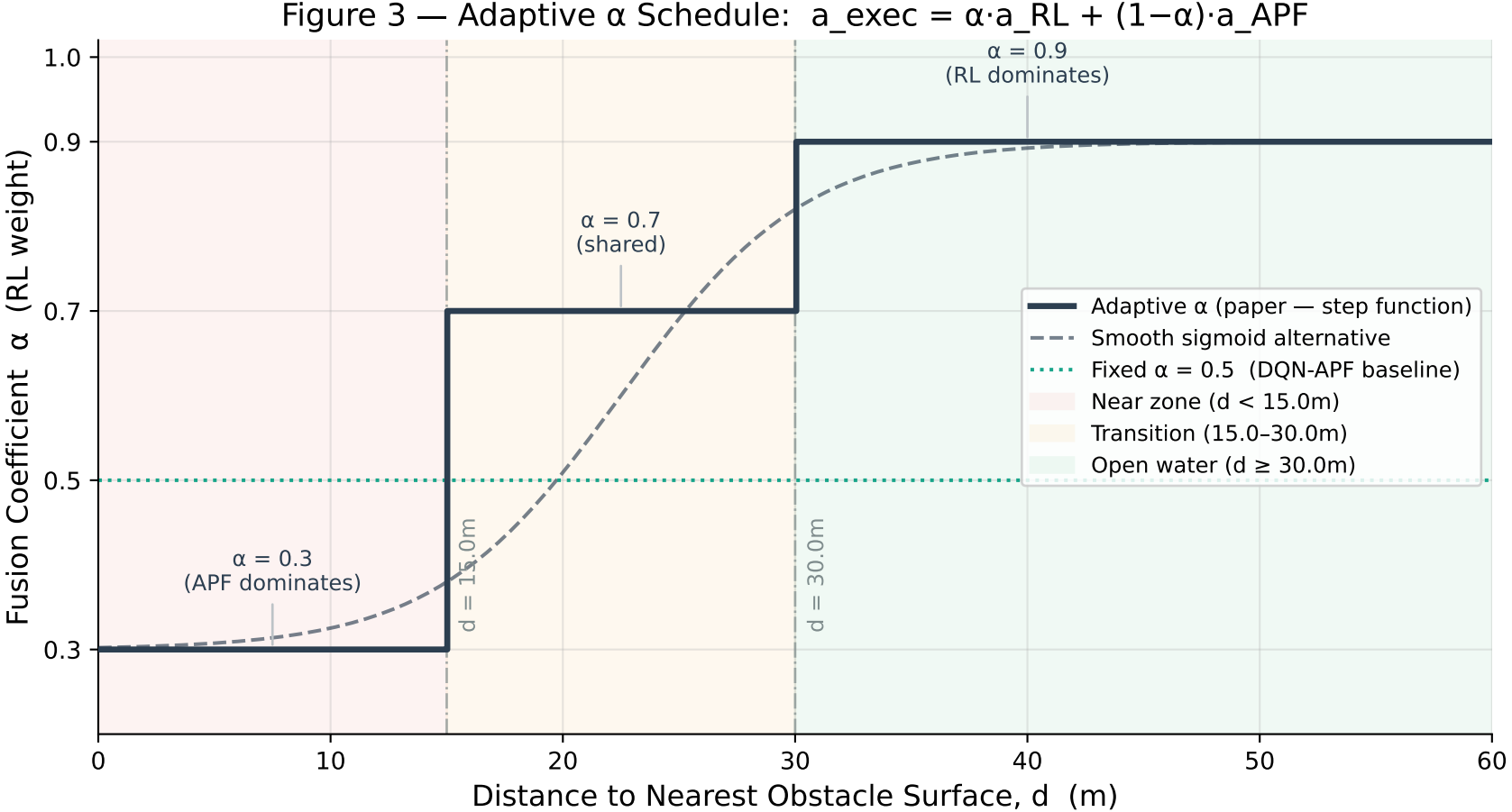
Figure 3 — Adaptive α Schedule: a_exec = α·a_RL + (1−α)·a_APF
Fusion Coefficient α (RL weight)
1.0
0.9
0.7
0.5
0.3
0
10
20
30
40
50
60
Distance to Nearest Obstacle Surface, d (m)
α = 0.3
(APF dominates)
α = 0.7
(shared)
α = 0.9
(RL dominates)
d = 15.0m
d = 30.0m
Adaptive α (paper — step function)
Smooth sigmoid alternative
Fixed α = 0.5 (DQN-APF baseline)
Near zone (d < 15.0m)
Transition (15.0–30.0m)
Open water (d ≥ 30.0m)